

Introduction to Operating System



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- Web References/MOOCs

➤ What is an Operating System?

- Bootstrap process
- Abstract view
- Why do we need an operating system?
- Computer System Components
- Types of OS and its features

- Textbook

- Abraham Silberschatz, Peter B. Galvin, Greg Gagne-Operating System Concepts, Wiley (2012).

- Reference books

- Ramez Elmasri, A Carrick, David Levine, Operating Systems, A Spiral Approach - McGrawHill Science Engineering Math (2009).
 - Remzi H. Arpaci-Dusseau, Andrea C. Arpaci-Dusseau, Operating Systems, Three Easy Pieces, Arpaci-Dusseau Books, Inc (2015).

- Web References/MOOCs

- <https://nptel.ac.in/courses/106/105/106105214/>
 - https://www.tutorialspoint.com/operating_system/index.htm
 - <https://www.javatpoint.com/os-tutorial>

Assessment Methods

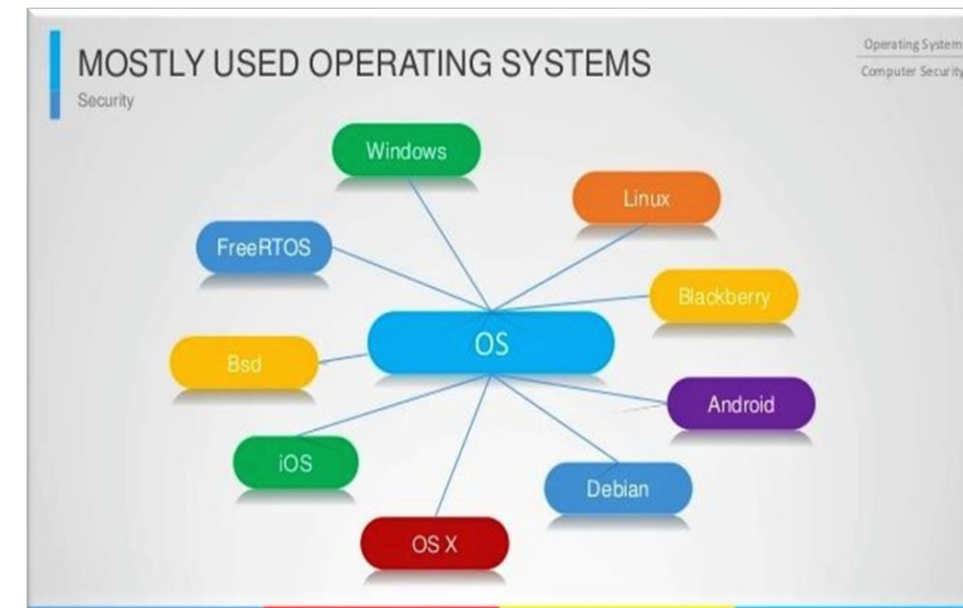
Assessment type	Max. Marks	Weightage
Assignment 1	10	10
Quiz 1	10	10
Quiz 2	10	10
CAT – I	50	15
CAT – II	50	15
FAT	100	40

Model 1: Introduction

- What is an Operating System?
- Mainframe Systems
- Desktop Systems
- Multiprocessor Systems
- Distributed Systems
- Clustered System
- Real -Time Systems
- Handheld Systems
- Feature Migration

What is an Operating System?

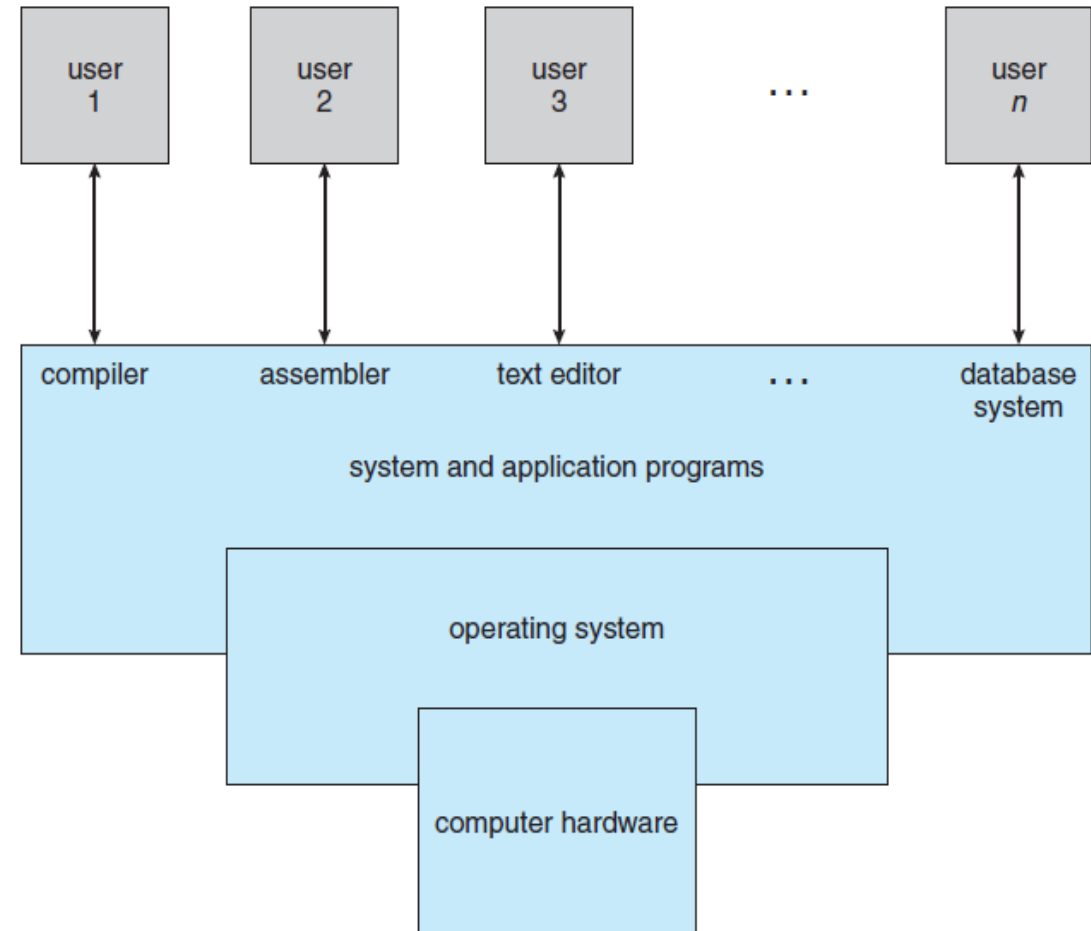
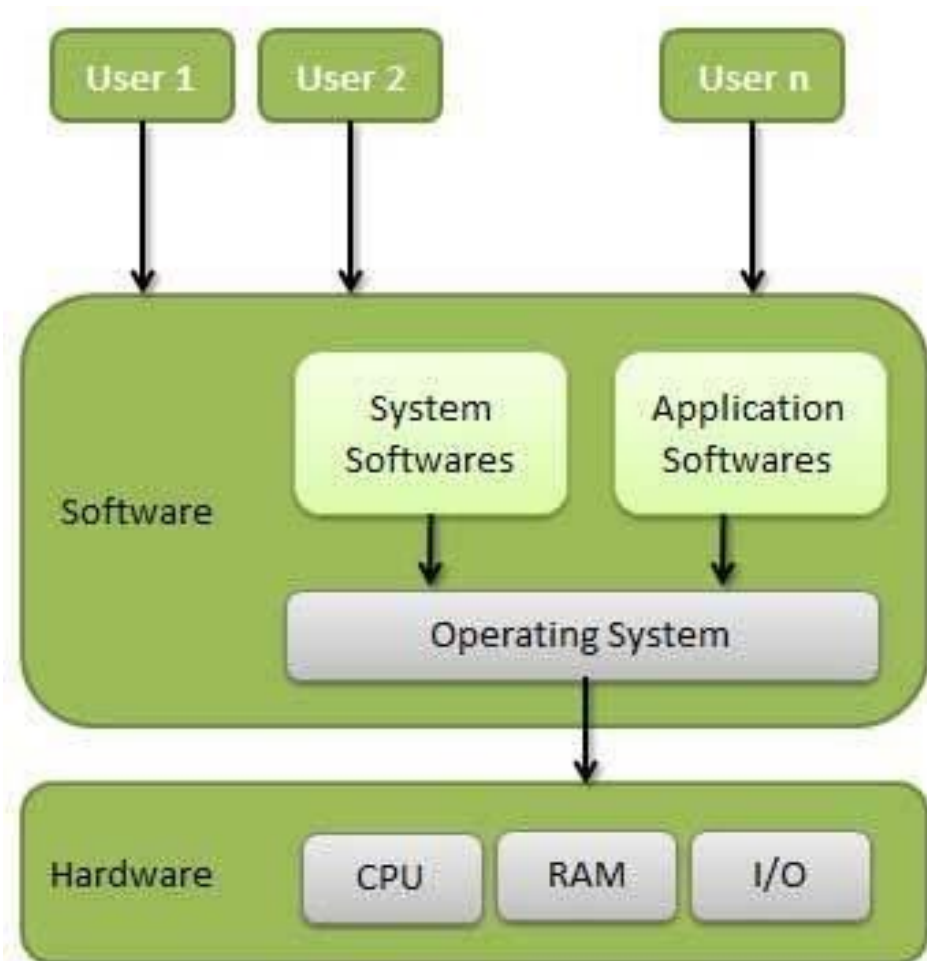
- An Operating System (OS) is a program that manages the computer hardware.
- A program that acts as an intermediary between a user of a computer and the computer hardware
- Operating system goals:
 - Execute user programs and make solving user problems easier
 - Make the computer system convenient to use
 - Use the computer hardware in an efficient manner



Computer System Components

- Computer system can be divided into four components:
 - Hardware – provides basic computing resources
 - CPU, memory, I/O devices
 - Operating system
 - Controls and coordinates use of hardware among various applications and users
 - Application programs – define the ways in which the system resources are used to solve the computing problems of the users
 - Word processors, compilers, web browsers, database systems, video games
 - Users
 - People, machines, other computers

Abstract View of System Components

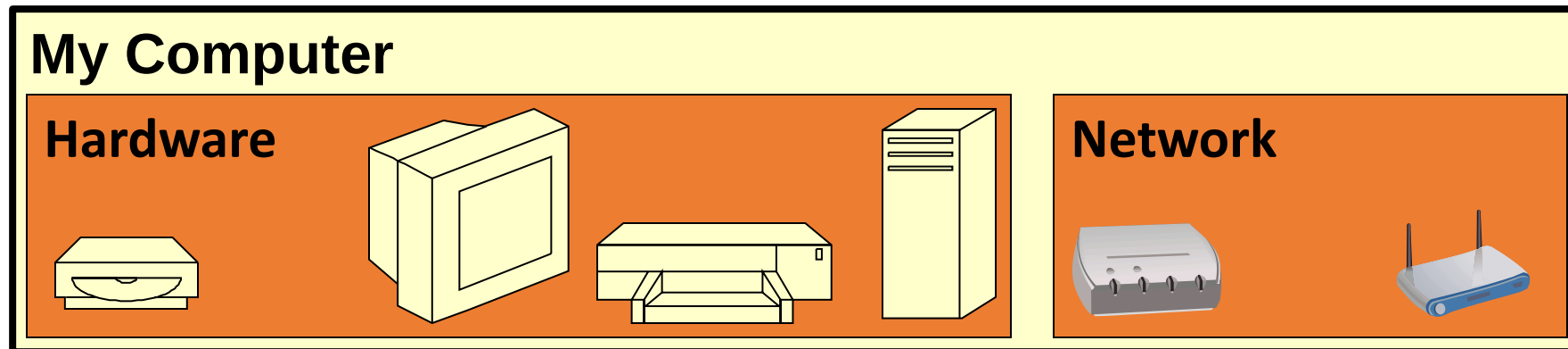


Goals of an Operating System

- Simplify the execution of user programs and make solving user problems easier.
- Use computer hardware efficiently.
- Allow sharing of hardware and software resources.
- Make application software portable and versatile.
- Provide security and protection among user programs.
- Improve overall system reliability.

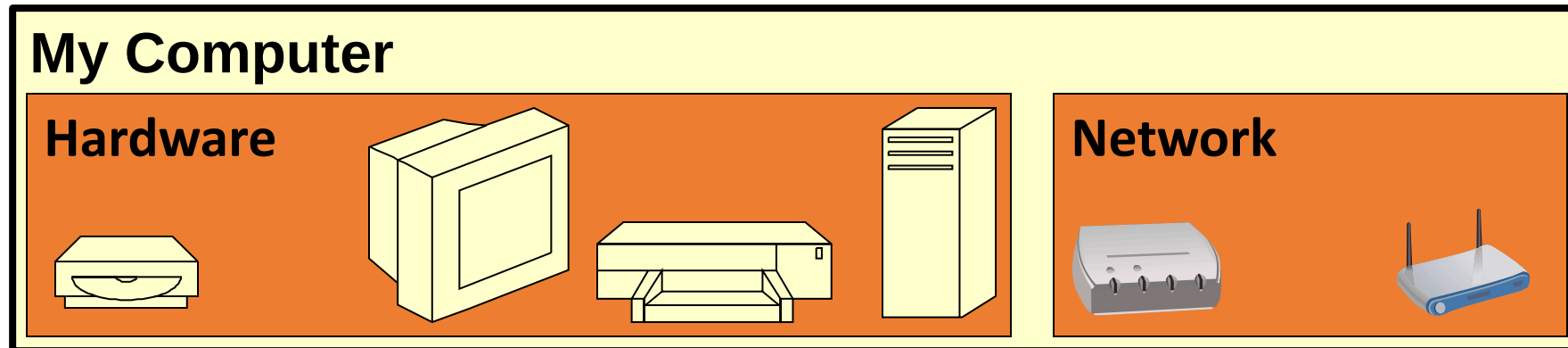
Why do we need an operating system?

- What do we have?
 - Set of common resources



Why do we need an operating system?

- What do we have?
 - Set of common resources
- What do we need?



Why do we need an operating system?

Application Software

Firefox

Second Life

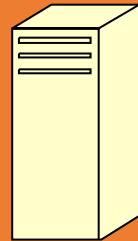
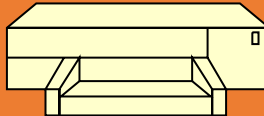
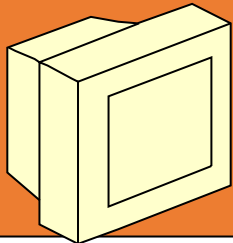
Yahoo
Chat

GMail

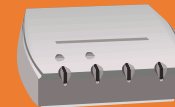
What do we have?

- Set of common resources
- What do we need?
 - A clean way to allow applications to use these resources!

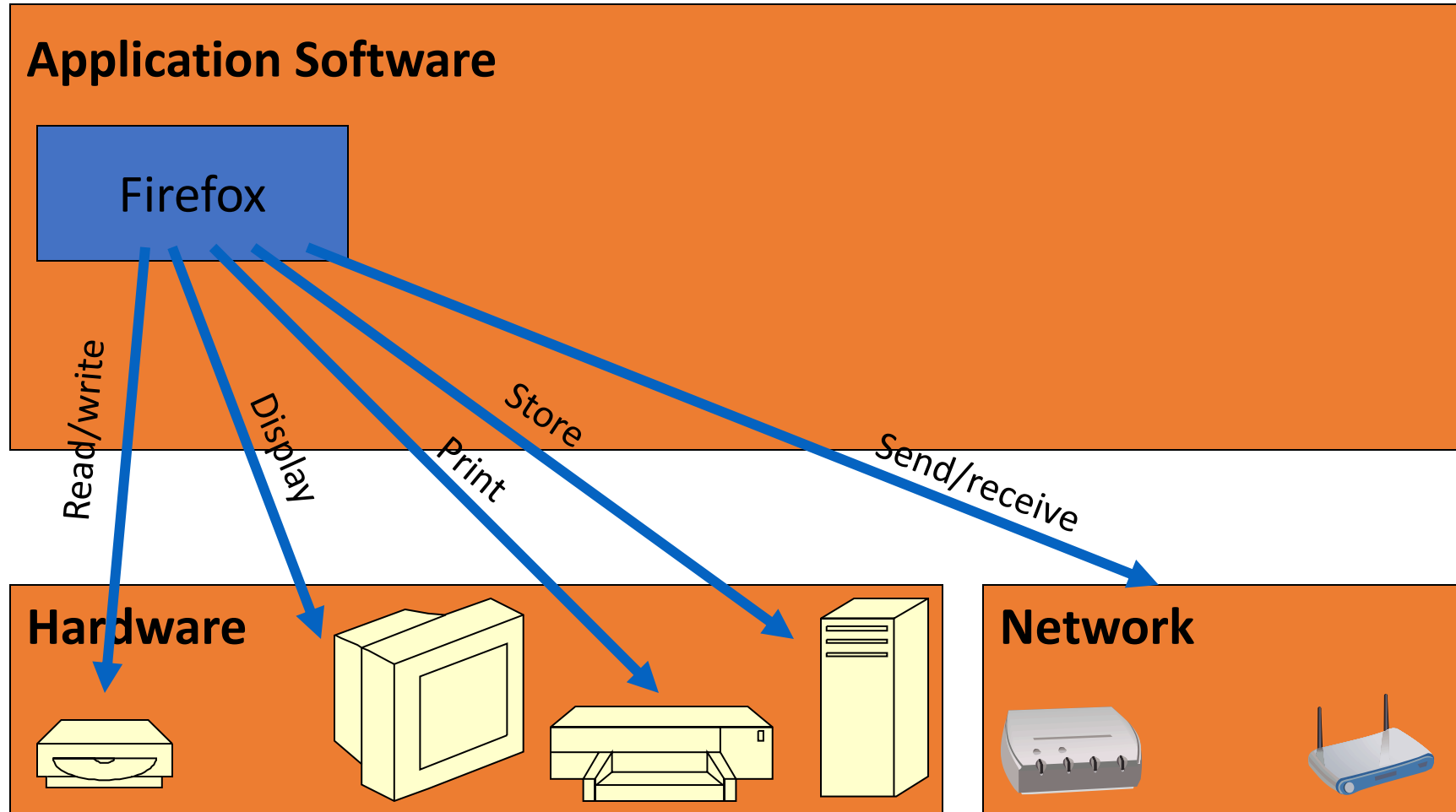
Hardware



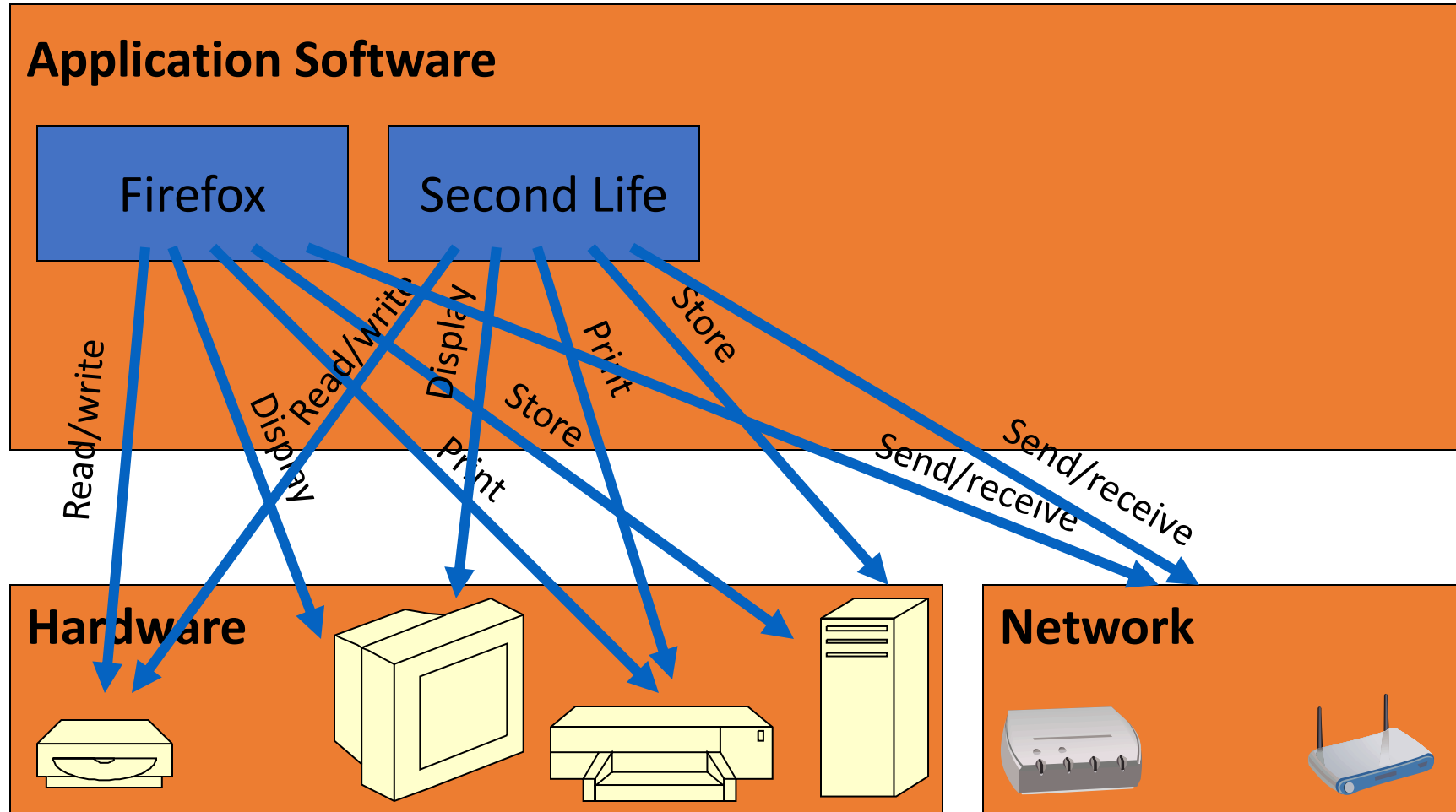
Network



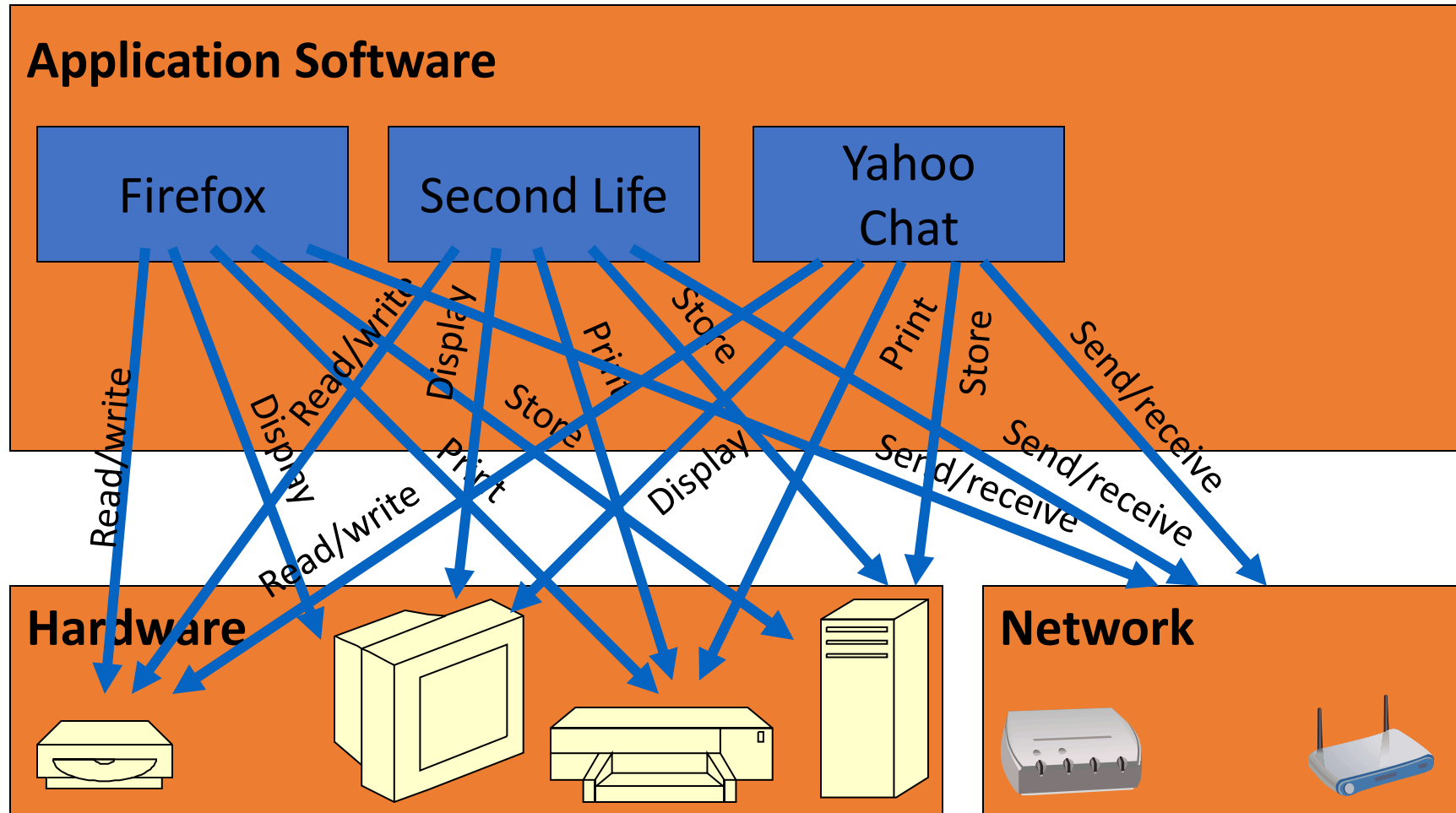
Application Requirements



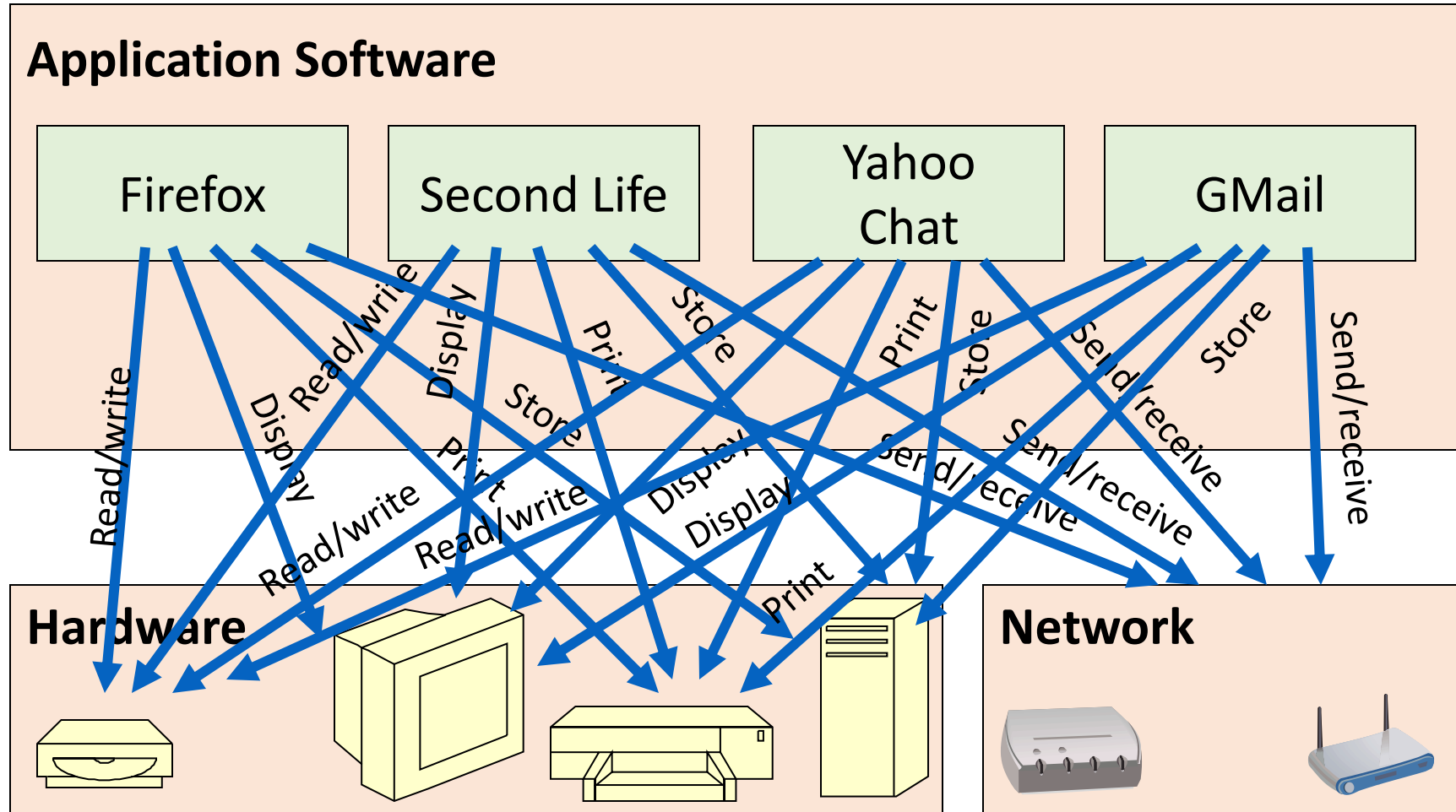
Two Applications?



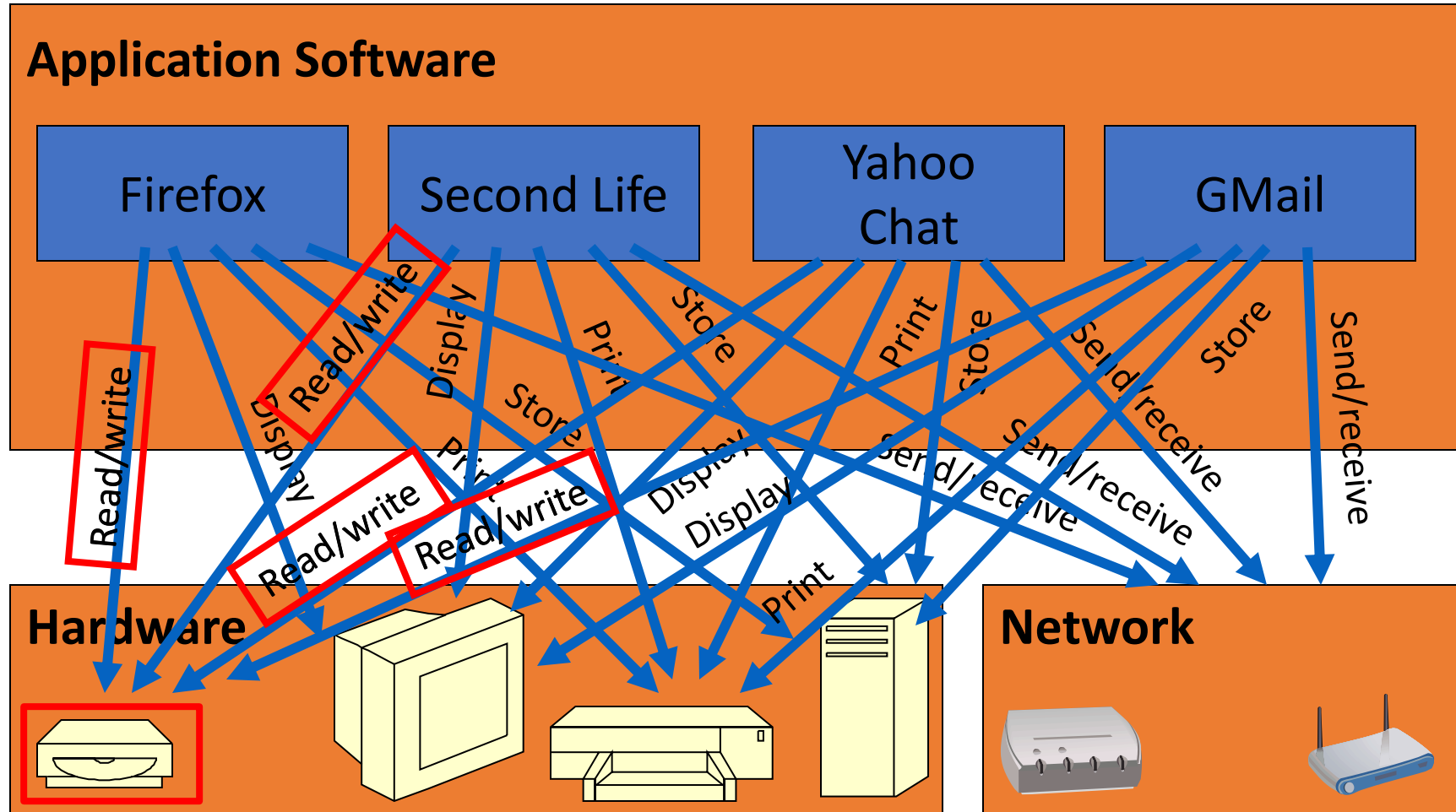
Managing More Applications?



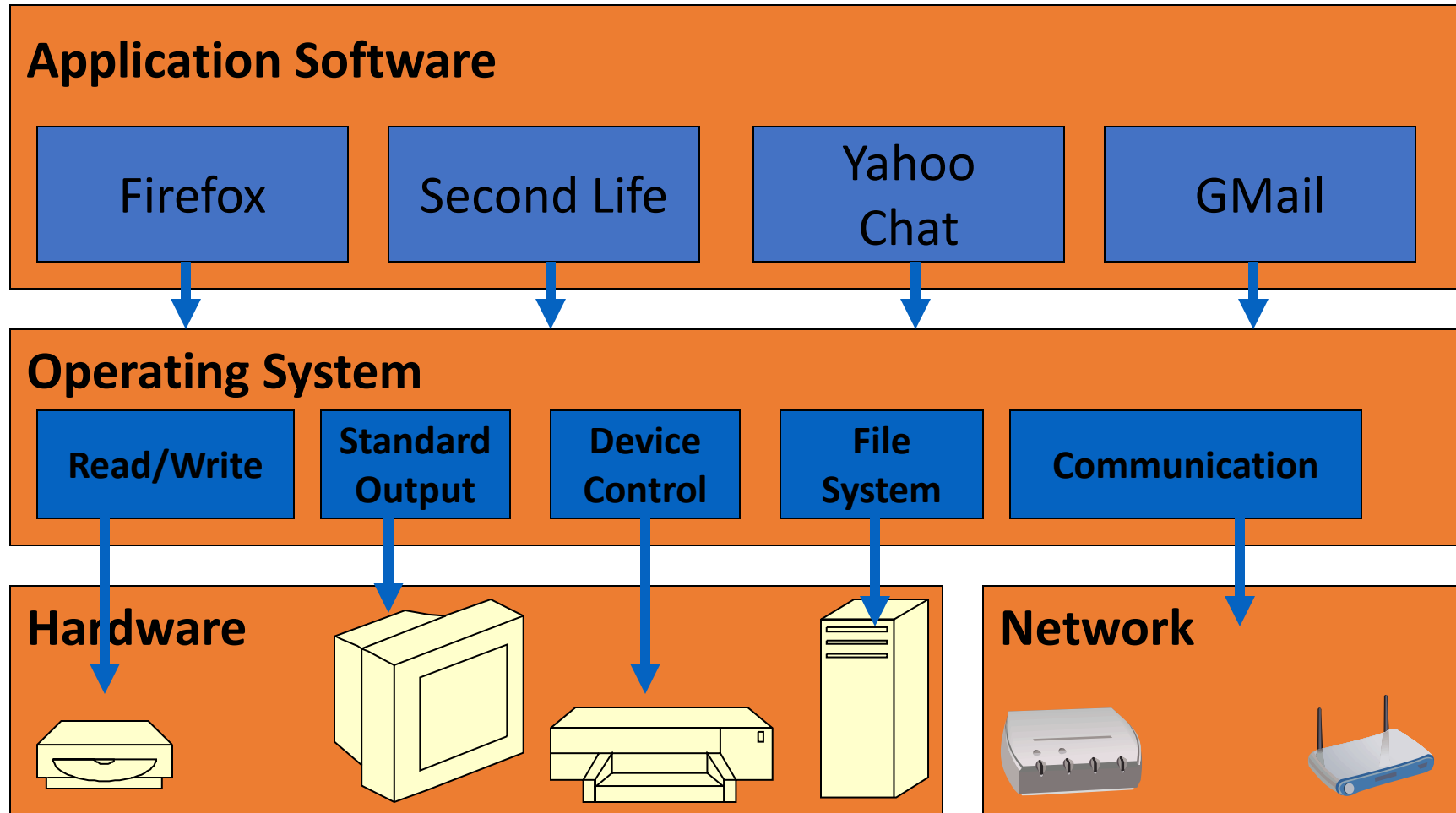
We need help!



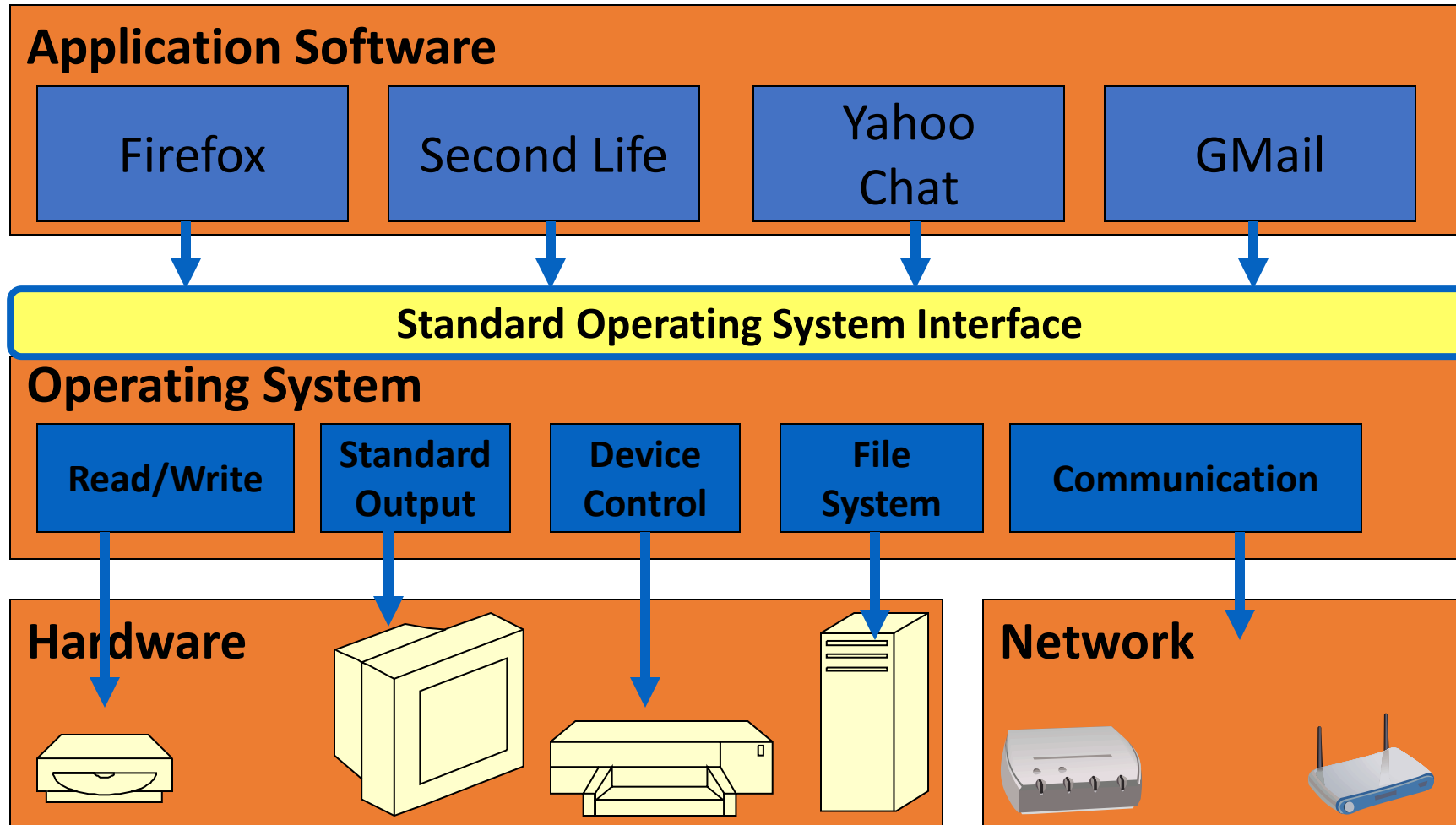
Approach: Find Common Functions



Delegate Common Functions



Export a Standard Interface



Operating System Definitions

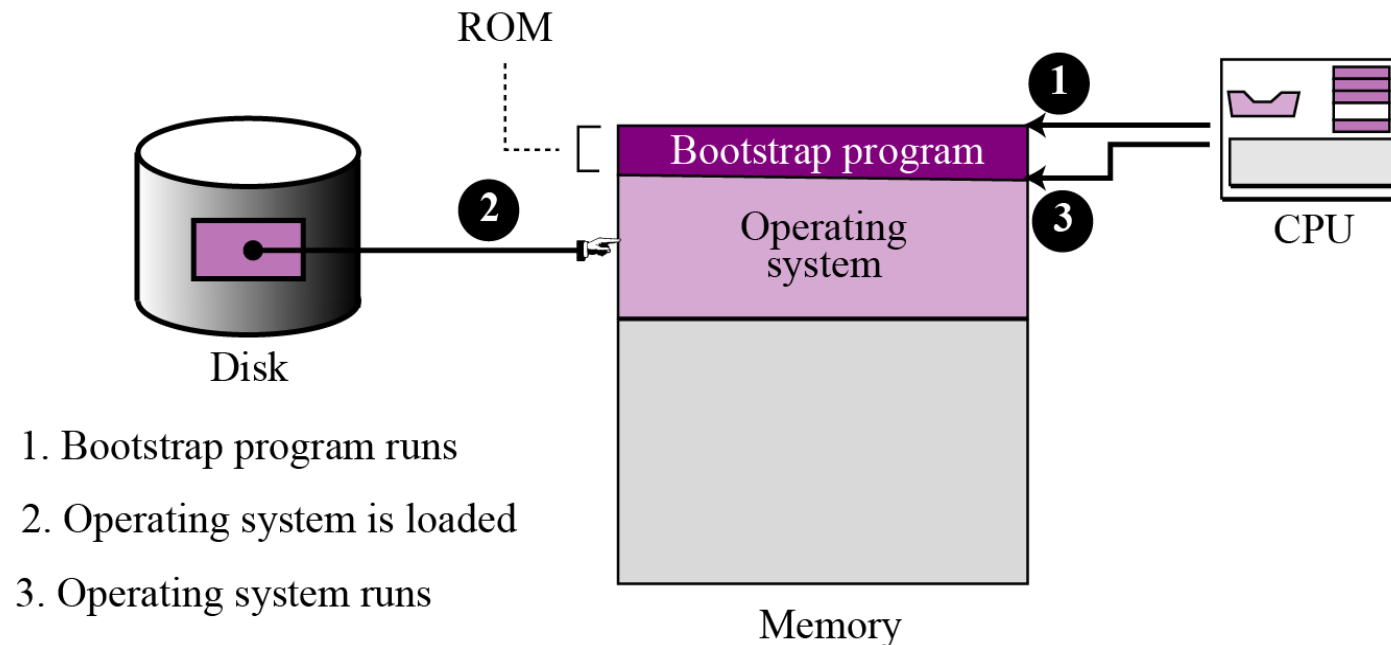
- OS is a **resource allocator**
 - Manages all resources
 - Decides between conflicting requests for efficient and fair resource use
- OS is a **control program**
 - Controls execution of programs to prevent errors and improper use of the computer

Operating System Definition (Cont.)

- No universally accepted definition
- “Everything a vendor ships when you order an operating system” is a good approximation
 - But varies wildly
- “The one program running at all times on the computer” is the **kernel**.
- Everything else is either
 - a system program (ships with the operating system) , or
 - an application program.

Computer Startup

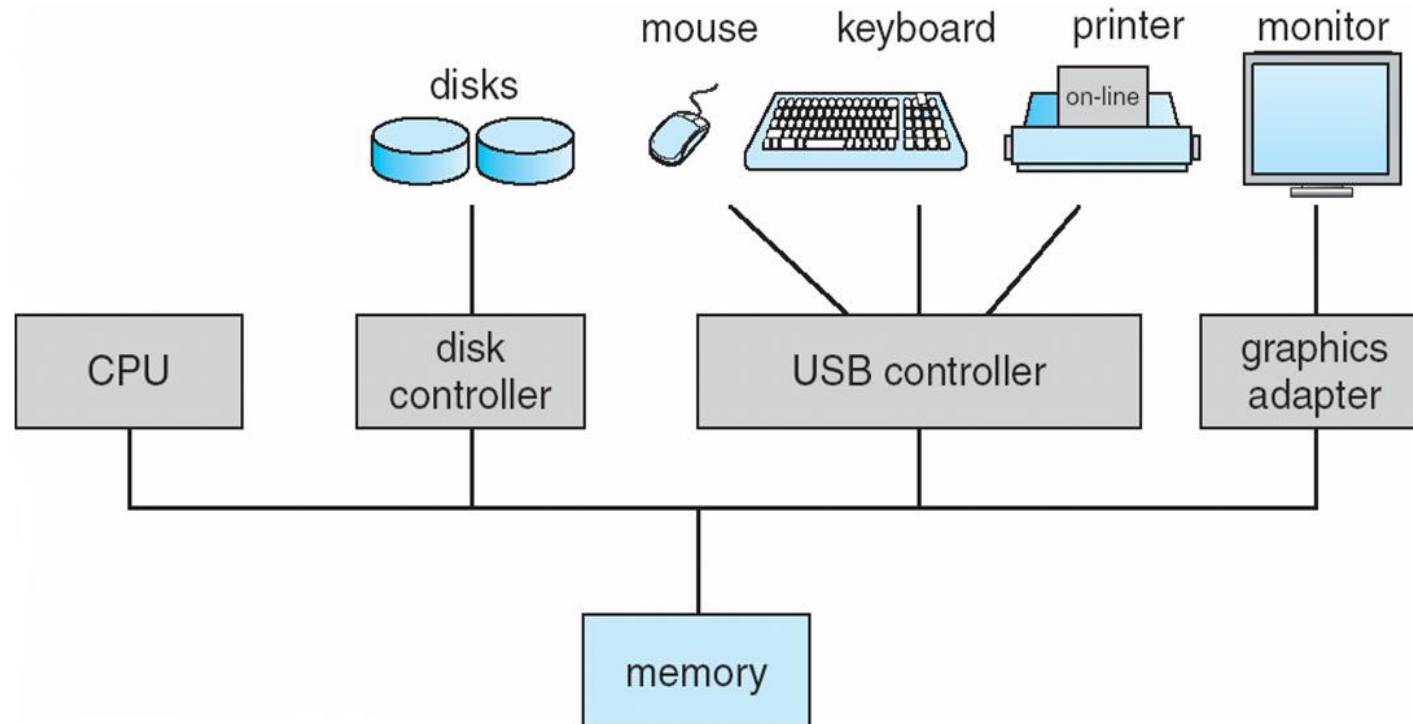
- **Bootstrap program** is loaded at power-up or reboot
 - Typically stored in ROM or EPROM, generally known as **firmware**
 - Initializes all aspects of system
 - Loads operating system kernel and starts execution



Computer System Organization

- Computer-system operation

- One or more CPUs, device controllers connect through common bus providing access to shared memory
- Concurrent execution of CPUs and devices competing for memory cycles



Mainframe Systems

- Computers used to tackle many commercial and scientific application.
- Reduce setup time by batching similar jobs
- Automatic job sequencing – automatically transfers control from one job to another. First rudimentary operating system
- Resident monitor
 - initial control in monitor
 - control transfers to job
 - when job completes control transfers back to monitor

Mainframe Systems (cont.)

- Growth of main frame system:-
 - Simple Batch system
 - Multiprogrammed System
 - Time-sharing System

Batch system

- User did not interact directly with computer system .
- User prepared a job which consisted of program, the data, and some control information and submitted to computer operator.
- OS task was to transfer control automatically from one job to another job.
- To speedup processing, operator batched similar jobs together and ran them.
- CPU is often idle.

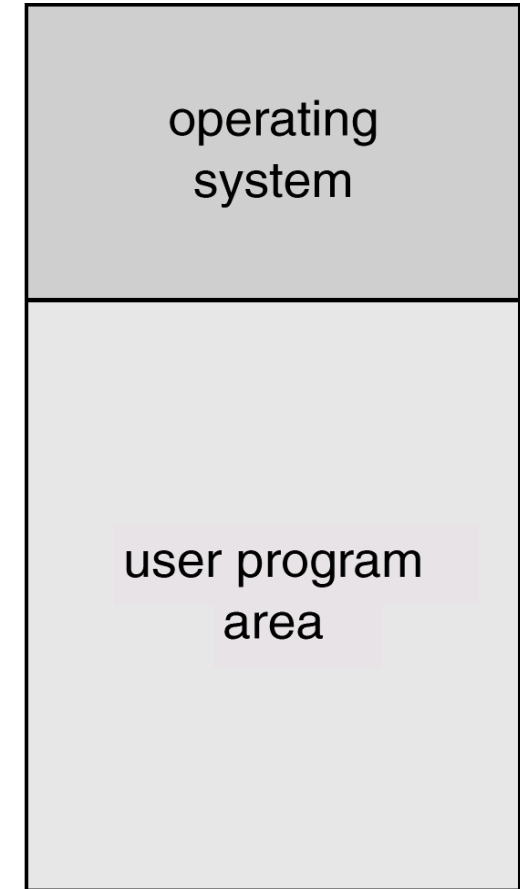
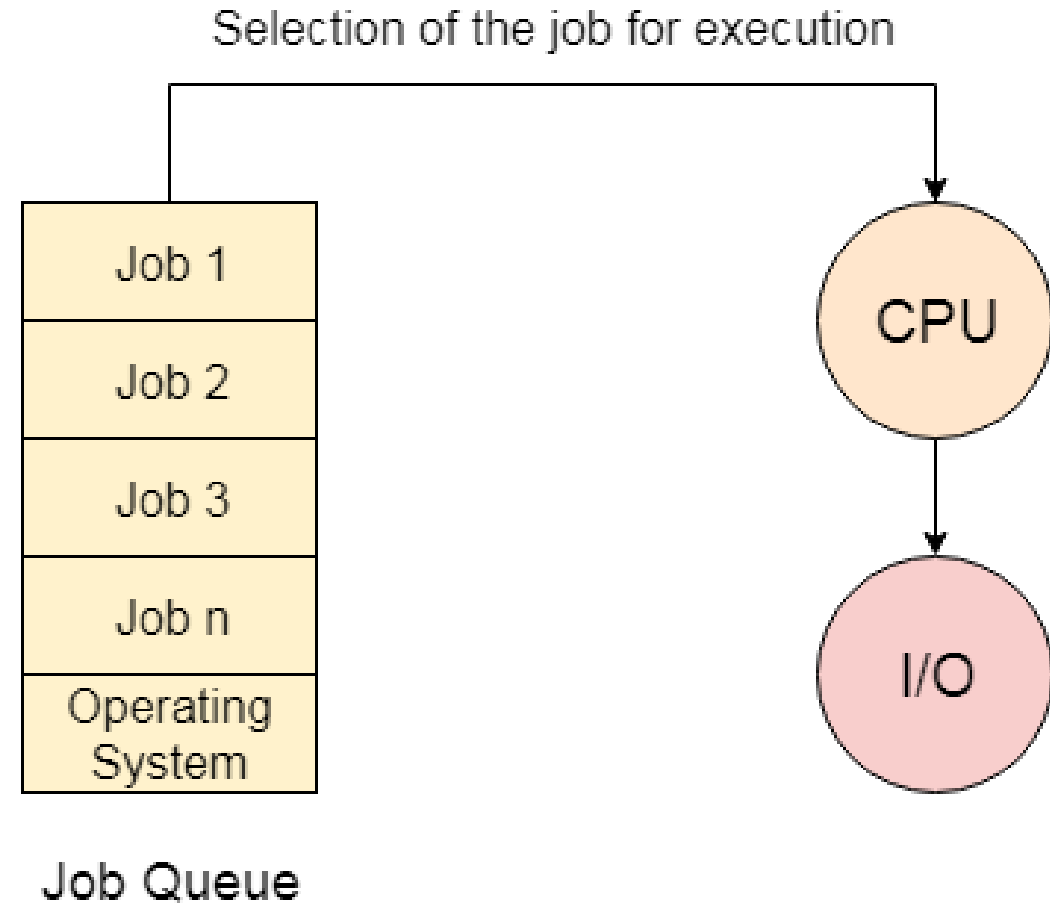


Figure: Memory Layout for a Simple Batch System

Batch system(cont.)

- The system put all of the jobs in a queue on the basis of first come first serve and then executes the jobs one by one. The users collect their respective output when all the jobs get executed.



Multiprogrammed System

- Several jobs are kept in main memory at the same time.
- OS picks and begin to execute one of the jobs in the memory.
- When job is waiting for some I/O operation the CPU is switched to another job and so on.
- Multiprogramming increases CPU utilization.
- Job Scheduling
- CPU Scheduling

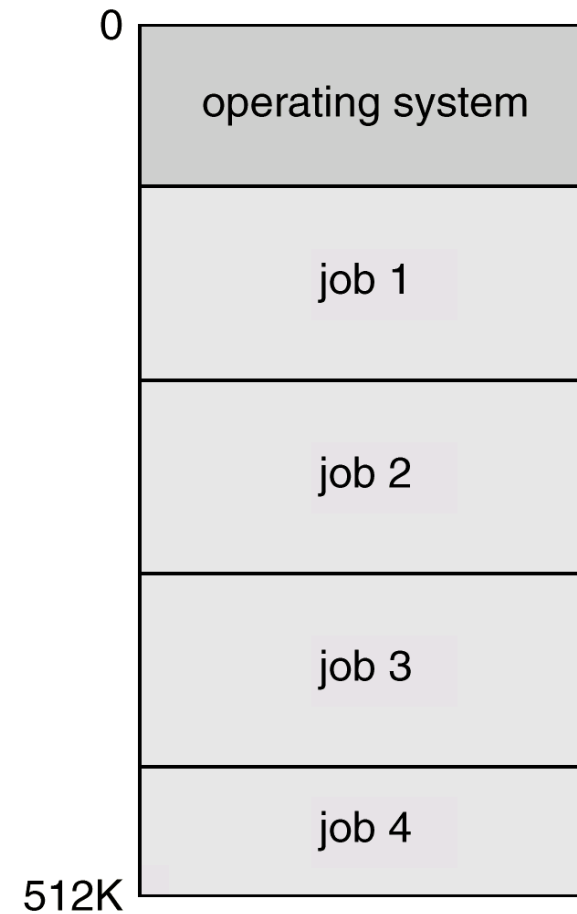
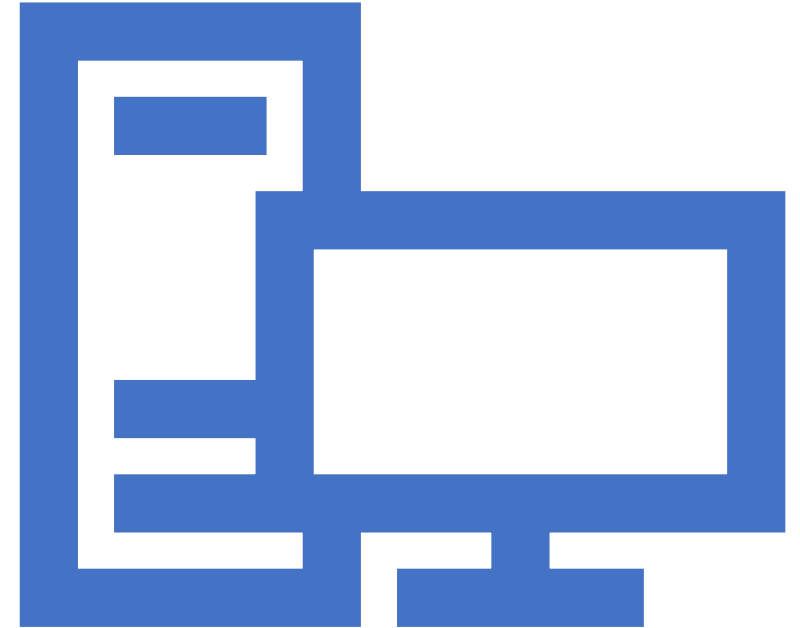


Figure: Memory Layout for a Multiprogramming System

OS Features Needed for Multiprogramming

- I/O routine supplied by the system
- Memory management – the system must allocate the memory to several jobs
- CPU scheduling – the system must choose among several jobs ready to run
- Allocation of devices



Time-sharing System

- A logical extension of multiprogramming
- CPU execute multiple jobs by switching among them, but the switches occur so frequently that users can interact with each program while it is running.
- Time-shared operating system allows many users to share the computer simultaneously.
- Time-shared OS uses CPU scheduling and multiprogramming to provide each user with a small portion of timed shared computer.
- A job swapped in and out of memory to the disk
- On-line communication between the user and the system is provided
 - When the operating system finishes the execution of one command, it seeks the next “control statement” from the user’s keyboard
- On-line system must be available for users to access data and code

Desktop Systems

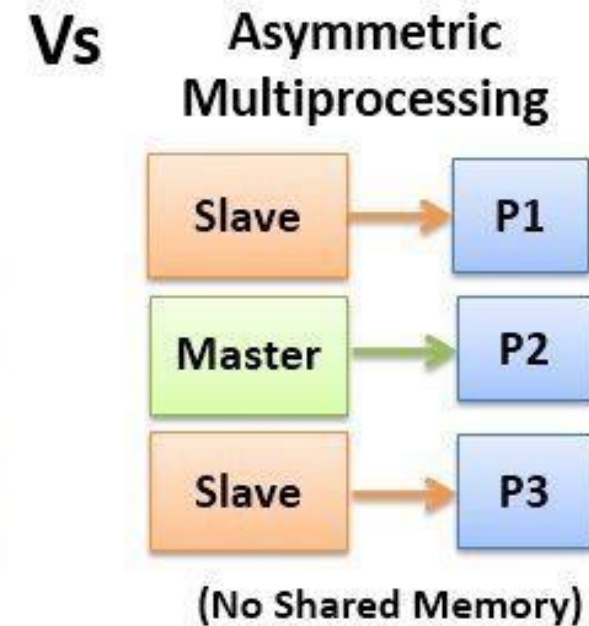
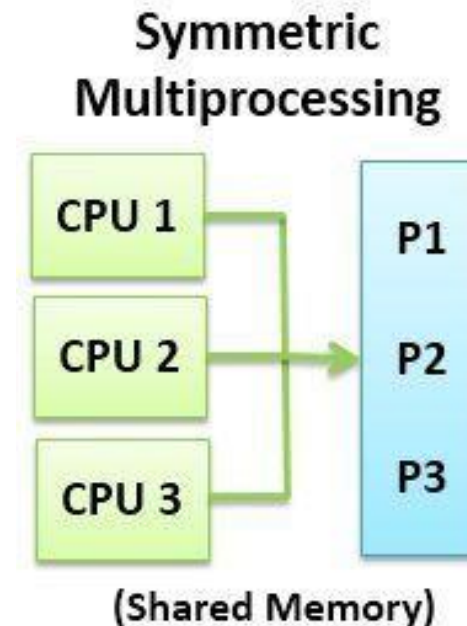
- Personal computers – computer system dedicated to a single user
- I/O devices – keyboards, mice, display screens, small printers
- User convenience and responsiveness
- Can adopt technology developed for larger operating system
 - Often individuals have sole use of computer and do not need advanced CPU utilization of protection features
- May run several different types of operating systems (Windows, MacOS, UNIX, Linux)

Multiprocessor System

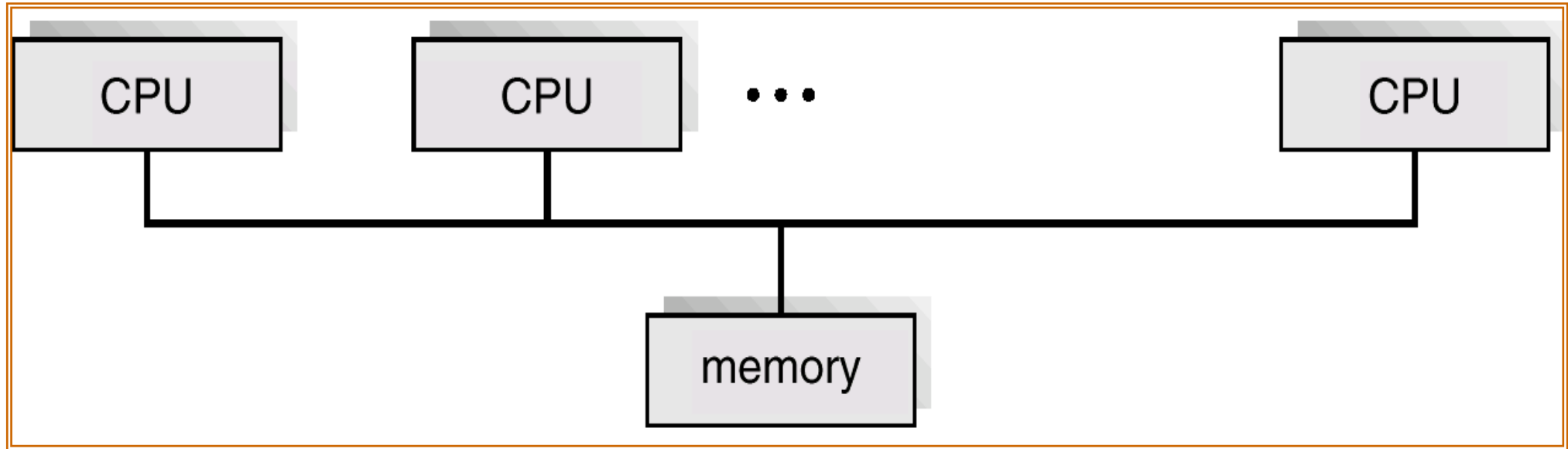
- also known as parallel system or tightly coupled system.
- Multiprocessor systems with more than one CPU in close communication sharing the computer bus, clock some time memory and peripheral devices.
- Multiprocessor System have three main advantage
 1. Increased throughput.
 2. Economy of scale.
 3. Increased reliability.

Multiprocessor System(cont.)

- Symmetric multiprocessing (SMP)
 - Each processor runs an identical copy of the operating system.
 - Copies communicate with another as needed.
 - Most modern operating systems support SMP.
- Asymmetric multiprocessing
 - Each processor is assigned a specific task;
 - master processor schedules and allocates work to slave processors.
 - More common in extremely large systems
 - Sun's operating system SunOS Ver 4 provides asymmetric multiprocessing.



Symmetric Multiprocessing Architecture



Distributed Systems

- Distribute the computation among several physical processors
- *Loosely coupled system* – each processor has its own local memory; processors communicate with one another through various communications lines, such as high-speed buses or telephone lines
- Advantages of distributed systems
 - Resources Sharing
 - Computation speed up – load sharing
 - Reliability
 - Communications

Distributed Systems(cont.)

- Requires networking infrastructure
- Local area networks (*LAN*) or Wide area networks (*WAN*)
- May be either *client-server* or *peer-to-peer* systems

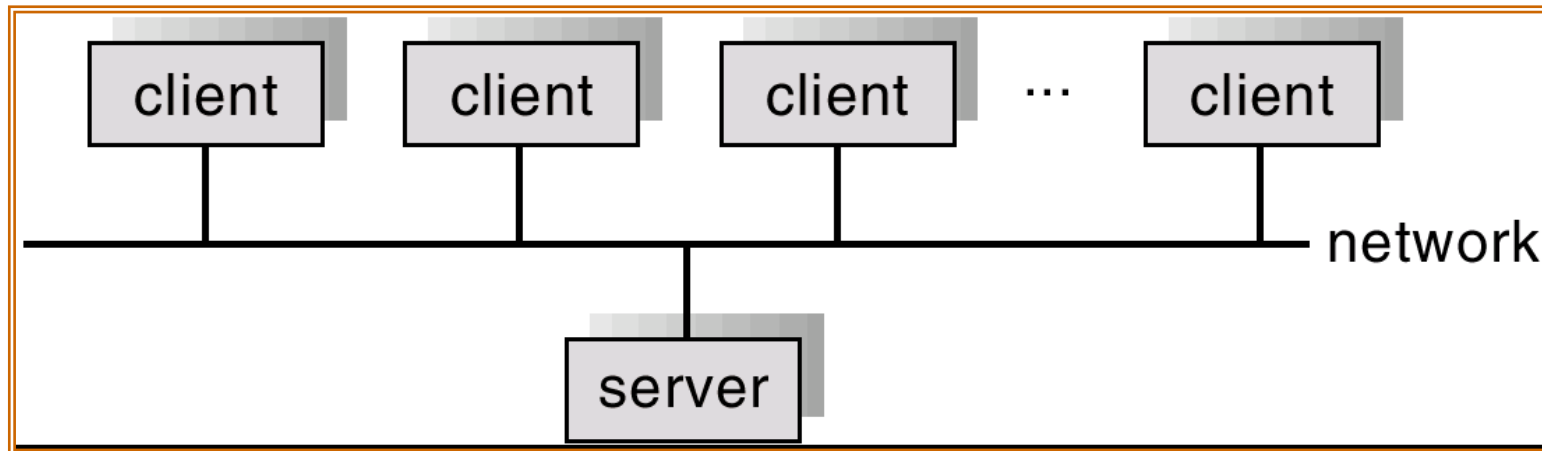


Figure: General Structure of Client-Server

Clustered Systems

- Clustering allows two or more systems to share storage.
- Clustering usually perform to provide high reliability.
- Asymmetric clustering: one server runs the application while other servers standby. The standby server does nothing but monitor the active server. If that server fails, the standby host becomes the active server.
- Symmetric clustering: all N hosts are running the application. This mode is more efficient, as it uses all of the available hardware.

Real-Time Systems

- Often used as a control device in a dedicated application such as controlling scientific experiments, medical imaging systems, industrial control systems, and some display systems
- Well-defined fixed-time constraints
- Real-Time systems may be either *hard* or *soft* real-time
- Processing must be done within the defined constraints otherwise system will fail.

Hard real time system

- Tasks have to be performed on time
- Failure to meet deadlines is fatal
- Example :

Flight Control System,
medical systems .

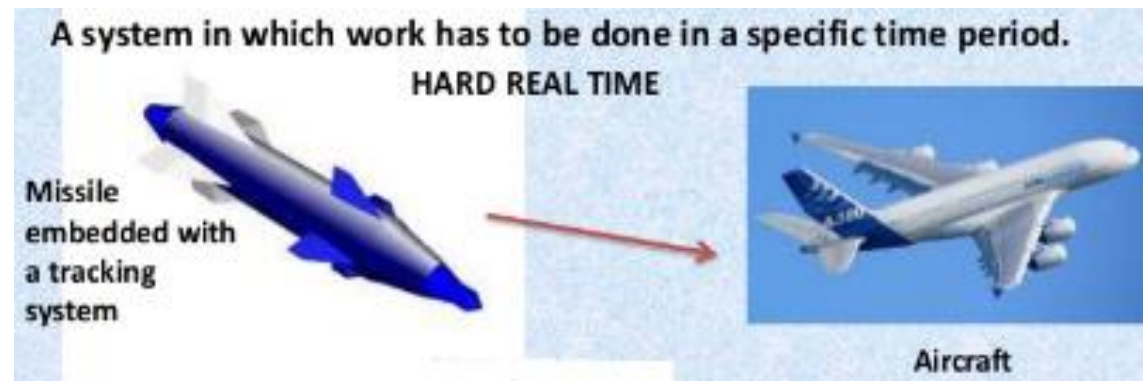
Soft real time system

- Tasks are performed as fast as possible
- Late completion of jobs is undesirable but not fatal.

• Example:
Online Databases,
computer games.

Real-Time Systems(cont.)

- Hard real-time:
 - It guarantees that critical task be completed on time.
 - Secondary storage limited or absent, data stored in short term memory, or read-only memory (ROM)
 - Conflicts with time-sharing systems, not supported by general-purpose operating systems
- Soft real-time
 - Limited utility in industrial control of robotics
 - Integrate-able with time-share systems
 - Useful in applications (multimedia, virtual reality) requiring tight response times



Handheld Systems

- It also known as personal digital assistants (PDAs)
- Nearly all modern PDAs have the ability to connect to the Internet.
 - Cellular telephones
- Issues:
 - Limited memory
 - Slow processors
 - Small display screens

Migration of Operating System Concepts and Features

